

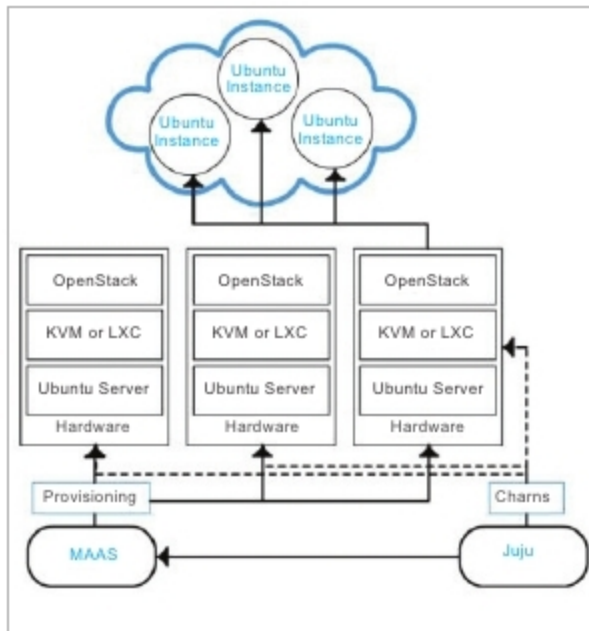


Figure 7: How Canonical Autopilot works

etc, can be performed on several machines simultaneously through an intuitive dashboard.

Canonical Autopilot makes use of MAAS for automated bare metal provisioning, Juju for deploying OpenStack Charms and Landscape for the administration, monitoring and management of the deployed OpenStack cluster. Besides these, advanced system monitoring tools like Nagios can be integrated with Autopilot using Landscape.

Figure 7 illustrates how Canonical Autopilot works.

Features

Listed below are the features of Canonical Autopilot:

- It has support for automated OpenStack deployment and administration using Ubuntu servers.
- It sets up clouds, adds new hardware to existing clouds and assists with cloud management.
- It has the ability to run OpenStack administrative services inside dedicated containers via LXD, i.e., Canonical's home-grown virtualisation hypervisor.

Official website: <http://www.ubuntu.org.cn/cloud/openstack/autopilot>

Compass

Compass is an open source tool for the automated deployment and management of OpenStack. It reduces complexity, saves

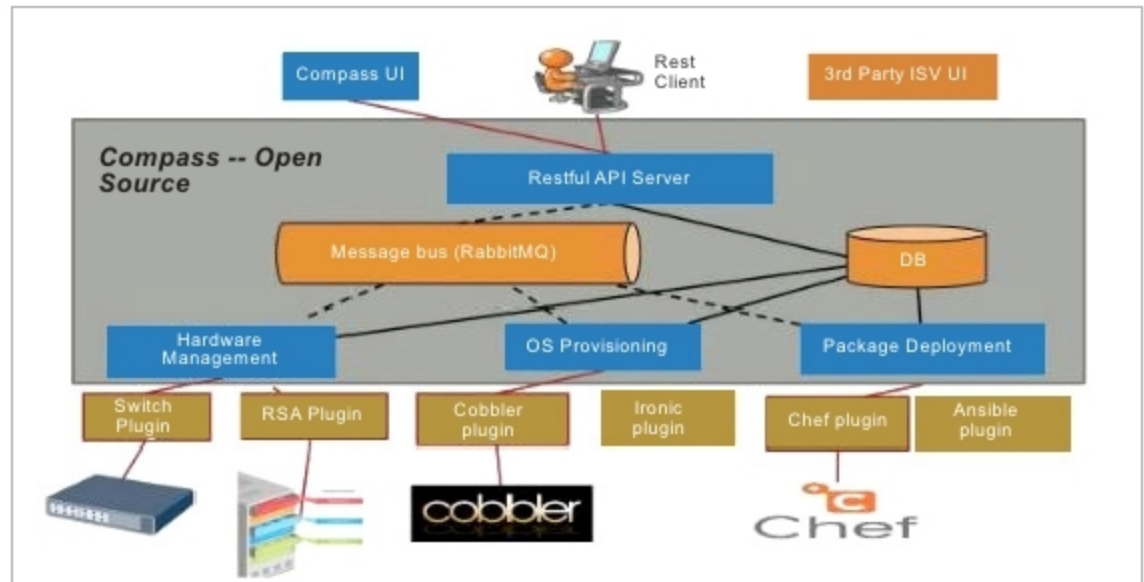


Figure 8: Compass' architecture

time and controls all sorts of errors in the data centre server management. It assists in bootstrapping the server pool associated with any cloud platform from bare metal nodes. It assists systems administrators in discovering hardware, deploying the OS and hypervisor, and also provides comprehensive configuration management.

Architecture

Figure 8 illustrates Compass' architecture.

Components

The components that make up Compass OpenStack software are listed below.

- **RESTful API server.**
- **Web UI:** This is written in AngularJS and used by developers.
- **Meta-Data module:** This allows a developer to extend the core functionality and provide a custom data model for OpenStack configurations — for example, with or without HA, single controller vs multi-node, etc.

- **Adapter interface:** This is used for automatic resource discovery.
- **Cobbler interface:** This is used for OS provisioning.

Features

Listed below are the features of Compass:

- Assists in the infrastructure bootstrapping process and also offers programmability for operators to do this.
- Allows implementation of different configuration flavours through metadata.
- Implements extensibility through the integration of a number of tools like Chef and Ansible for OpenStack cluster configuration. By default, Ansible is used for OpenStack installation. The Compass core blends with other tools for resource discovery, OS provisioning and package deployment.

Official website: <https://wiki.openstack.org/wiki/Compass> 

References

- [1] <https://www.chef.io/chef/>
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- [3] <https://jujucharms.com/>
- [4] <https://puppet.com>
- [5] <https://launchpad.net/fuel>
- [6] <http://www.ubuntu.org.cn/cloud/openstack/autopilot>
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